Lecture 4 – Lab 4

Tree & Substitution method

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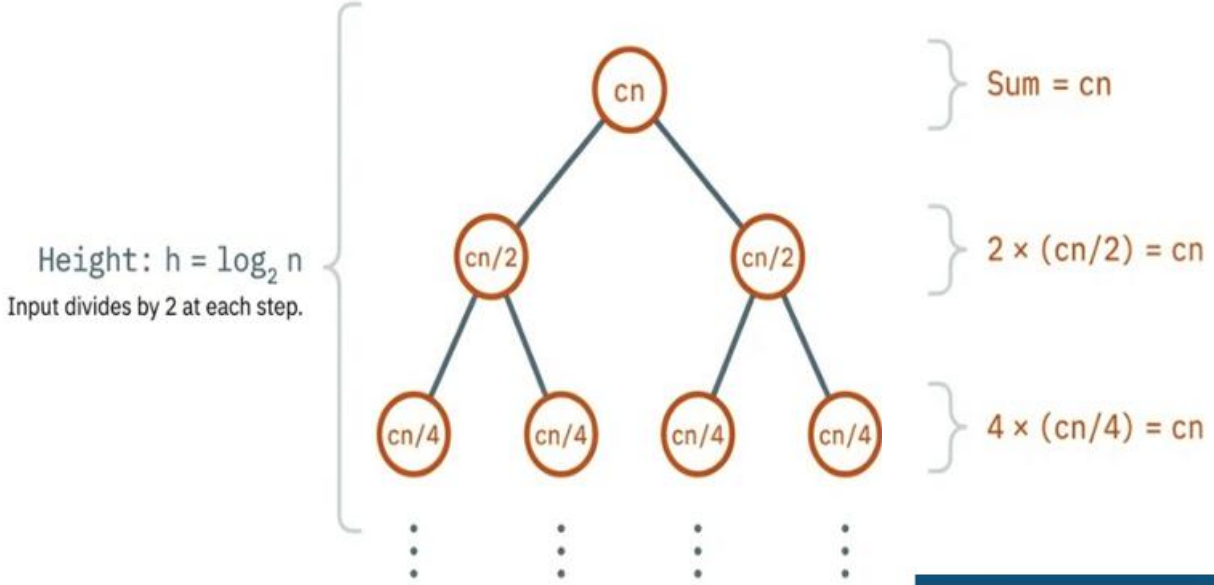


Summation Logic

How to add the levels based on their relationship

Common Summation Patterns

Constant Series

$$\text{Sum} = \frac{n - n}{n}$$

Arithmetic Series

$$\text{Sum} = n(n+1)(2n+1)$$

Quadratic Series

(The Square Case)

$$\text{Sum} = \frac{n(n+1)(2n+1)}{6}$$

Geometric Series

Decreasing ($r < 1$, Root Dominates):

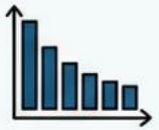
$$\frac{a}{1-r}$$

Increasing ($r > 1$, Leaves Dominate):

$$a\left(\frac{(1-r^h)}{1-r}\right)$$

Harmonic Series

(The Reciprocal Case)



$$\text{Sum} \approx \ln(n)$$

Questions

TASK: SOLVE USING THE TREE METHOD

01. $T(n) = 2T(n/2) + 10$

02. $T(n) = 3T(n/3) + n^2$

03. $T(n) = T(n-1) + T(n-2) + 1$

04. $T(n) = T(n-1) + n$

05. $T(n) = 3T(n/2) + n$

06. $T(n) = T(n/4) + T(n/2) + n^2$

07. $T(n) = T(n/3) + T(2n/3) + n$

08. $T(n) = 2T(n/4) + n/2$

09. $T(n) = 3T(n/4) + n$

10. $T(n) = T(n-1) + 1$

11. $T(n) = T(n-1) + 1/n$

12. $T(n) = T(n-1) + (n-1)^2$

Solve using substitution method

$$T(n) = T(n-1) + 1$$

$$T(n) = 2T(n/4) + \sqrt{n}$$